PROJECT PRESENTATION

Small vs Large Language Model

Which one is more accurate, and which one uses less energy?

Task: Natural Language Inference (SNLI) | Models: DeBERTa vs Mistral-7B

What we did

We compared a **small** AI model with a **large** AI model on one task. We checked two things: **how often each is correct** and **how much energy each uses**.

The task: read two sentences and decide how they are related.

Sentence 1	Sentence 2	Answer
A man is playing football.	A person is playing a sport.	**Entailment** (must be true)
A man is playing football.	The man is asleep.	**Contradiction** (cannot be true)
A man is playing football.	The man is a professional.	**Neutral** (maybe true)

Data: SNLI, a standard collection of sentence pairs with the correct answers already known.

How it was done: the steps in the code

1

Get the data

Loaded SNLI test data, removed unclear pairs, and picked 1,000 pairs at random.

2

Load the two models

DeBERTa (small) and Mistral-7B (large) on a Google Colab GPU.

3

Get answers

Both models got the same 1,000 pairs and answered with one of the three labels.

4

Check and measure

Compared answers with the correct ones. Also measured time and GPU energy.

5

Repeat

Ran everything 5 times for each model and took the average.

The two models

DeBERTa-v3-base (small)

- 184 million parameters
- Already trained to do this exact task
- Reads both sentences together
- Gives the answer directly
- Memory used: 0.69 GB

Mistral-7B (large)

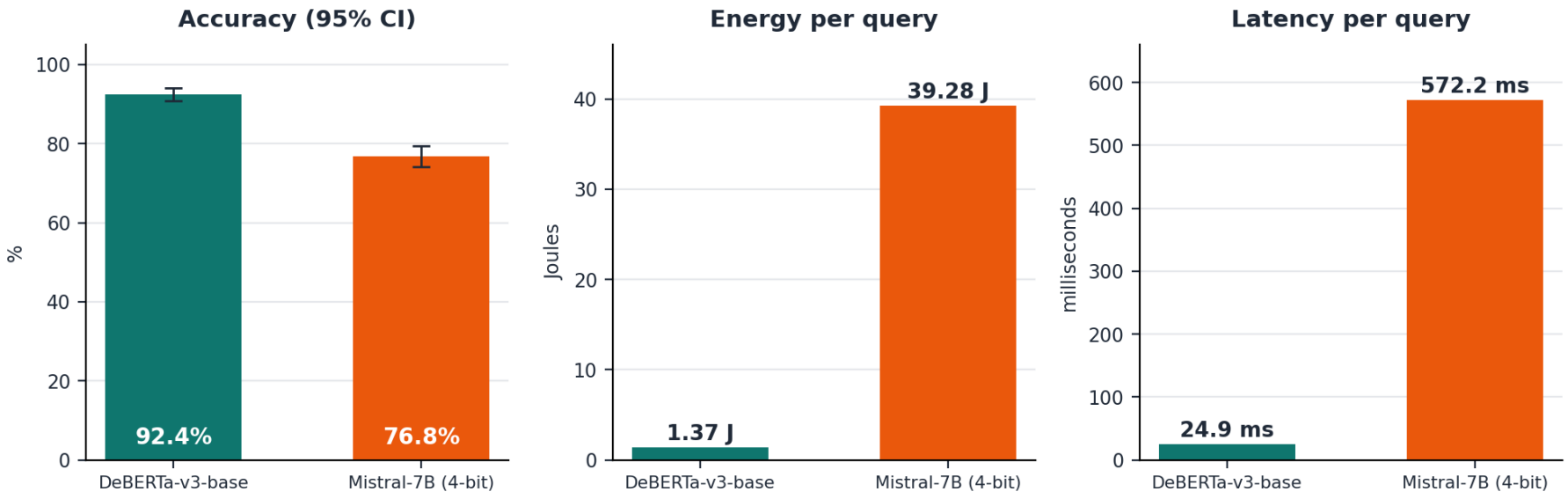
- 7 billion parameters
- A general-purpose model, not made for this task
- We gave it short instructions and 6 examples
- It answers Yes, No or Maybe
- Compressed to 4-bit to fit the GPU. Memory used: 3.74 GB

How we measured energy

- The GPU has a **built-in energy counter**. We read it before and after each run.
- Energy used = counter after minus counter before.
- Energy per query = total energy divided by 1,000 pairs.
- We ran a few **warm-up** queries first, so start-up time is not counted.
- We repeated the full run **5 times** to make sure the numbers are stable.
- We also recorded the **time** each query took.

This is a real measurement, not an estimate.

Results



The small model **wins on all three**: more accurate, faster, and much less energy.

Results in numbers

	DeBERTa (small)	Mistral-7B (large)
Accuracy	**92.4%**	76.8%
Time per query	**25 ms**	572 ms
Energy per query	**1.37 J**	39.3 J
CO_2 per query*	**0.00027 g**	0.00764 g

- The small model is **15.6 points more accurate**.
- Mistral uses about **29 times more energy** per query.
- Mistral is about **23 times slower**.

Based on 1,000 random SNLI test pairs, average of 5 runs. * Using 0.7 kg CO2 per kWh of electricity.

CONCLUSION

For this task, the small model gave **better answers** while using about **29 times less energy**.